

An Embedded Virtual Experiment Environment System for Reality Classroom

YanXiang Zhang*

YuTong Zi†

JiaYu Wang‡

University of Science and Technology of China

ABSTRACT

We designed a low-cost augmented virtuality system based on the Oculus Quest to embed VR in classrooms. To build the system, we measure the size and position of tables in the classroom, make a proxy model in Unity, and then embed the proxy model to seamlessly within the real classroom. In this system, schoolchildren can realize collaborative experiments in ideal conditions or some hard-to-reach scenes. This system's contribution is: (1) By manually adding obstacles, it makes up for most VR systems that can only delimit the area but cannot identify obstacles. (2) It cleverly reuses tables and makes them play the role of anti-collision, workbench, and joystick placement. (3) It expands the available area of VR in complex environments.

Keywords: Augmented virtuality, anti-collision, classroom.

Index Terms: H.5.1 [Information Interfaces and Presentation]: Multimedia Information Systems—Artificial, Augmented, and Virtual Realities; K.3.1 [Computers and Education]: Computer Uses in Education

1 INTRODUCTION

Augmented Virtuality is the mutual enhancement between the real object and the virtual scene by integrating the whole experience into the virtual environment [1]. It can allow users to interact with fictional and real objects [2].

Experiential learning theory shows learners must directly participate in the environment because knowledge primarily comes from natural objects' impressions [3]. So, classroom education is always essential and irreplaceable. But for schoolchildren, many Gedanken experiments are challenging to carry out in real classrooms, like experiencing the different gravity on different planets, simulating a vacuum environment or a frictionless environment. So are those experiments with high costs and safety risks.

Many existing systems try to solve this problem, but we still found some unsolved core problems, for example, in 3D Virtual Worlds, a persistent virtual world [4]. It can bring new opportunities for cooperative learning, immersion, experimentation, and interaction but cannot support offline classrooms' teaching. Educational projects in 3D Virtual Learning Environments and 3D Virtual Worlds always have difficulties monitoring, storing, and processing students' interactions [5]. AR Learn system can provide highly flexible support for different educational settings [2]. Still, smartphones' immersion is not as good as VR, and this system does not suit schoolchildren because

smartphones are often not allowed in the classroom. Some AR systems need users to have the ability to program and model, and this may be too hard for schoolchildren [6]. VR systems can only roughly recognize the environment but cannot intelligently recognize obstacles. So, it always has problems in the classroom when tables divide the space.

Therefore, our research problems are achieving high-immersion collaborative teaching for school children in classrooms and extending the available VR area to avoid collisions. We build an Augmented virtuality system based on the real classroom. By manually adding obstacles, it can expand the available area and avoid collisions. Moreover, we can reuse tables to be workbenches in virtuality.

2 SYSTEM DESIGN

2.1 Establish a Proxy Model for the Reality Classroom

The first step is to measure the length, width, and height of the classroom and tables, and each table's relative position in the classroom. The classroom we use is 6.3m*6.3m. There are six tables inside. Each table is 1.2m long and 0.6m wide, and 0.65m high. We establish a rectangular coordinate system with the lower-left corner in front of the classroom as the origin. Then we build 1:1 proxy models of the classroom and tables in Maya. To reduce collisions and enhance the system's fault tolerance, we enlarge the table's proxy model's length and width by 2 cm. According to the real classroom, we place the proxy models in Unity.

2.2 Match and Fit the Proxy Model to the Real Space

We set 4 recognition points on each sidewall of the real classroom, and they are 1.5m away from the floor. It can help nest the classroom proxy model with the real one because we write a program to let the Oculus Quest identifies it. Then we need to calibrate the position of the obstacle. We paste different recognition points on the upper left and lower right corners of each table. Only when the real environment points coincide with the virtual one will the table's proxy model appear in the system. If not, the system will present a gray-scale realistic environment with some red spots to remind users to move the real table to the position it should be. After calibration, the users can start the game. To avoid the potential hazard caused by users moving a table, we fix calibrated tables with vacuum suction cups and nylon ties.

2.3 Proxy Models for Anti-collision

Proxy models for tables can remind users of their range of movement because they separate the classroom into small areas for each person. It also meets the class rules because students can only sit behind their tables during class. Through these models, users can also know other users' locations. By controlling the distance between tables, we can avoid collisions between people. We also slightly expand the volume of the proxy models of tables and fix the real tables to the floor with vacuum suction cups to maintain users' safety.

* email: petrel@ustc.edu.cn

† email: yvtongzi@mail.ustc.edu.cn

‡ email: jiayu0909@mail.ustc.edu.cn

2.4 Three game mode

The system provides three game modes. The first is a user-centered single-player mode. In this mode, proxy models of other tables will disappear so it can provide a broad view. According to different experiments, the user's table's proxy model can change its skins. It suits experiments under hard-to-reach scenes like space or ideal conditions like the vacuum and no friction. The second is single-player with multi-data mode. This mode adds a database on the single-player ones to suit experiments that can finish alone but always results in variety. The third one is the multiplayer collaborative mode. In this mode, users can see other tables and complete a task collaboratively. For example, to assemble high-precision models of rocket and drone. These high-precision models are expensive and easily damaged, so they don't suit schoolchildren's learning. The virtual model can reduce costs and avoid loss problem.

3 EXPERIMENT

We post our volunteer recruitment notice in a primary school. We promise that volunteers who participate can get \$20 in compensation, and we invite their guardians to accompany them during the experiment. Finally, we recruited 12 volunteers, 7 girls, and 5 boys. All of them are students in the fifth grade of primary school. As some collaborative projects in our test, we divide all volunteers into 2 groups. Group A: 3 girls and 3 boys. Group B: 4 girls and 2 boys. In the experiment, we will observe the performance of our volunteers. After completing the test, volunteers need to fill out a questionnaire that includes questions about collaborative feelings, willingness to continue using, ease of use, and learning effects.

In the classroom, volunteers need to experience three games. The first one is a single-player game involving gravity on different planets. In this game, volunteers can see the surrounding environment become the moon's surface, and an astronaut stands in front of them. Their tables became the spacecraft's dashboard, with jump command buttons, planet switching buttons, and a monitor showing the astronaut's jump height. The user makes the astronaut jump through the joystick. The second is a multi-data single-player game showing chemical processes from the molecular level. Volunteers can see the spherical molecular model of O₂ and C. Different mixing ratios will synthesize different substances. To observe others' results, volunteers can find that when O₂ is sufficient, there is only CO₂. If O₂ is insufficient, there are both CO and CO₂. The third one is a multiplayer collaborative game. Volunteers need to assemble a drone collaboratively. The figure below shows the classroom we use, a game we use, and a user perspective screenshot. Not installed parts are shown in red.



Figure 1: Test our system in a classroom.

4 RESULTS

In the experiment, volunteers are very focused, and no collision happens. 10 volunteers say the system is exciting and easy to use. 12 volunteers say this is the first time they can assemble a drone and experience chemical experiments. These kinds of experiments are hard to realize in a real classroom because of the high cost or

have a specific danger. They thought that sharing their experimental results with friends and assemble drones in teamwork was exciting. Collaboration seems efficient because most volunteers think they can understand and remember the knowledge faster than usual. 7 volunteers were delighted to recommend this system to their friends. All our volunteers hope their school can introduce our system so they can use it every day. They also gave us advice like saying the classroom model can be more realistic, and the game can be more complex.

5 DISCUSSION AND FUTURE WORK

Our augmented virtuality system solves some problems in the VR system. Most of the VR equipment can only roughly identify the environment, so users need to delimit the safety zone by themselves. In some complex environments with many obstacles, like classrooms, the available area for VR is too small, and collisions always happen. By manually adding obstacles, our system reuses obstacles, makes them play the role of anti-collision, workbench, and joystick placement, and expands the available area.

Our system is more immersive than AR teaching, and students do not need to hold mobile devices. Our system supports multiple students and allows teachers to explain during the experiment. Volunteers say the system has made up for the shortcomings of real classrooms and laboratories. Besides, the equipment we use and the upcoming Oculus Quest2 are both cheap, at US\$299.00, which supports the possibility that our system can be scaled to meet the needs of the community.

The weak point is the system's scalability needs to expand. So, for future work, we plan to add two functions. The first one is to allow users to personalize the obstacle model. We will put some standard obstacle models in the system. When users select the type of obstacle, they can input the actual size of their obstacle to establish personalized ones. The second is to add a real-time tracking function of obstacle position changes. Tracking technology can update real objects' position changes so that suction cups are no longer needed to fix the tables.

ACKNOWLEDGEMENT

The work is supported by Ministry of Education (China) Humanities and Social Sciences Research Foundation under Grant No.: 19A10358002.

REFERENCES

- [1] Hughes, C. E., & Stapleton, C. B. (2005, July). The shared imagination: Creative collaboration in augmented virtuality. In *HCI International* (pp. 22-27).
- [2] Ternier, S., Klemke, R., Kalz, M., Van Ulzen, P., & Specht, M. (2012). ARLearn: Augmented Reality Meets Augmented Virtuality. *J. UCS*, 18(15), 2143-2164.
- [3] M. B. Ibanez, D. Villaran, and C. Delgado-Kloos, "Integrating assessment into augmented reality-based learning environments," in *Proc. IEEE 15th Int. Conf. Advanced Learning Technologies*, Hualien, Taiwan, Jul. 6-9, 2015, pp. 218-222.
- [4] Chen, A., Golparvar-Fard, M., & Kleiner, B. (2013). SAVES: A safety training augmented virtuality environment for construction hazard recognition and severity identification. *CONVR*, 2013, 373-84.
- [5] J. Martin-Gutierrez and M. D. M. Fernandez, "Applying augmented reality in engineering education to improve academic performance & student motivation," *Int. J. Eng. Educ.*, vol. 20, no. 1, pp. 623-635.
- [6] Wei, H., Liu, Y., & Wang, Y. (2019, March). Building AR-based Optical Experiment Applications in a VR Course. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)* (pp. 1225-1226). IEEE.